

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I R. Rupasree (192472102) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Rupasree

Level - 4

HP

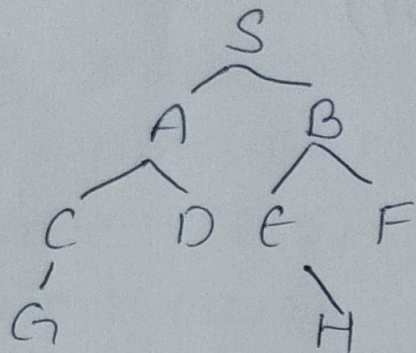
RUBRICS FOR CALCULATION:

Criteria	Weight	Excellent	Good	Fair	Poor
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1] Depth-Limited Search (DLS):-

To perform Depth-limited Search (DLS) with a depth limit of 3 and determine whether the goal can be reached.

Eg:-



Depth limit = 3

Levels:-

Level 0: S

Level 1: A B

Level 2: C D E F

Level 3: G H

Assume left to Right Expansion

Nodes Expanded: $S \rightarrow A \rightarrow C \rightarrow G$ (Goal found)

If Goal Lies beyond Depth limit:

Suppose the Goal is at Level 4

DLS stops expanding after level 3.

Result:

> Goal is not found.

> Algorithm cutoff, indicating search stopped because of depth limit, not because goal does not exist.

Result:-

Depth Limited Search explores nodes only up to specified depth. If goal is beyond limit, it returns cutoff's.

2. Missionaries and Cannibals Problem:-

To safely move 3 Missionaries and 3 Cannibals across the river.

Formulation:-

State: $(M\text{-left}, C\text{-left}, \text{Boat})$

[Boat $\rightarrow$ left or right]

eg:- $(3, 3, L)$

Initial State: $(3, 3, L)$

Goal State: $(0, 0, R)$

Action:-

Boat can carry:

- > 1 Missionary
- > 2 Missionary
- > 1 Cannibal
- > 2 Cannibal
- > 1 Missionary + 1 Cannibal

Transition Model:-

Move selected passengers across river while ensuring:

- > Boat carries at most 2 people
- > Missionaries are never outnumbered by cannibals on either bank.

Goal Test: $(0, 0, R)$

Initial: $(3, 3, L)$

After moving two cannibals: $(3, 1, R)$

One cannibal Returns: $(3, 2, L)$

Result:-

Breadth-first Search explores state level by level and guarantees the shortest valid solution.

3) Water Jug Problem:-

To measure use exactly 4 gallons of water using a 5-gallon jug and 3-gallon jug.

Formulation:-

State: Represented as (x, y) where

> x = amount of water in 5-gallon jug

> y = amount of water in 3-gallon jug

Initial State: $(0, 0)$

Goal State: $(4, x)$

Algorithm:-

- 1) Fill 5-gallon jug & pour water in 3-gallon jug
- 2) Empty 3-gallon jug
- 3) Transfer remaining water
- 4) Fill 5-gallon jug again
- 5) Pour water from 5-gallon jug into 3-gallon jug
- 6) Remaining water in 5-gallon jug is exactly 4 gallons.

Sequence of state:-

$(0, 0)$

$(5, 0)$

$(2, 3)$

$(2, 0)$

$(0, 2)$

$(5, 2)$

$(4, 3)$

The 5-gallon jug contains exactly 4 gallons of water.

Result:-

Exactly 4 gallons of water are measured successfully in 5-gallon jug using 5-gallon and 3-gallon jug.